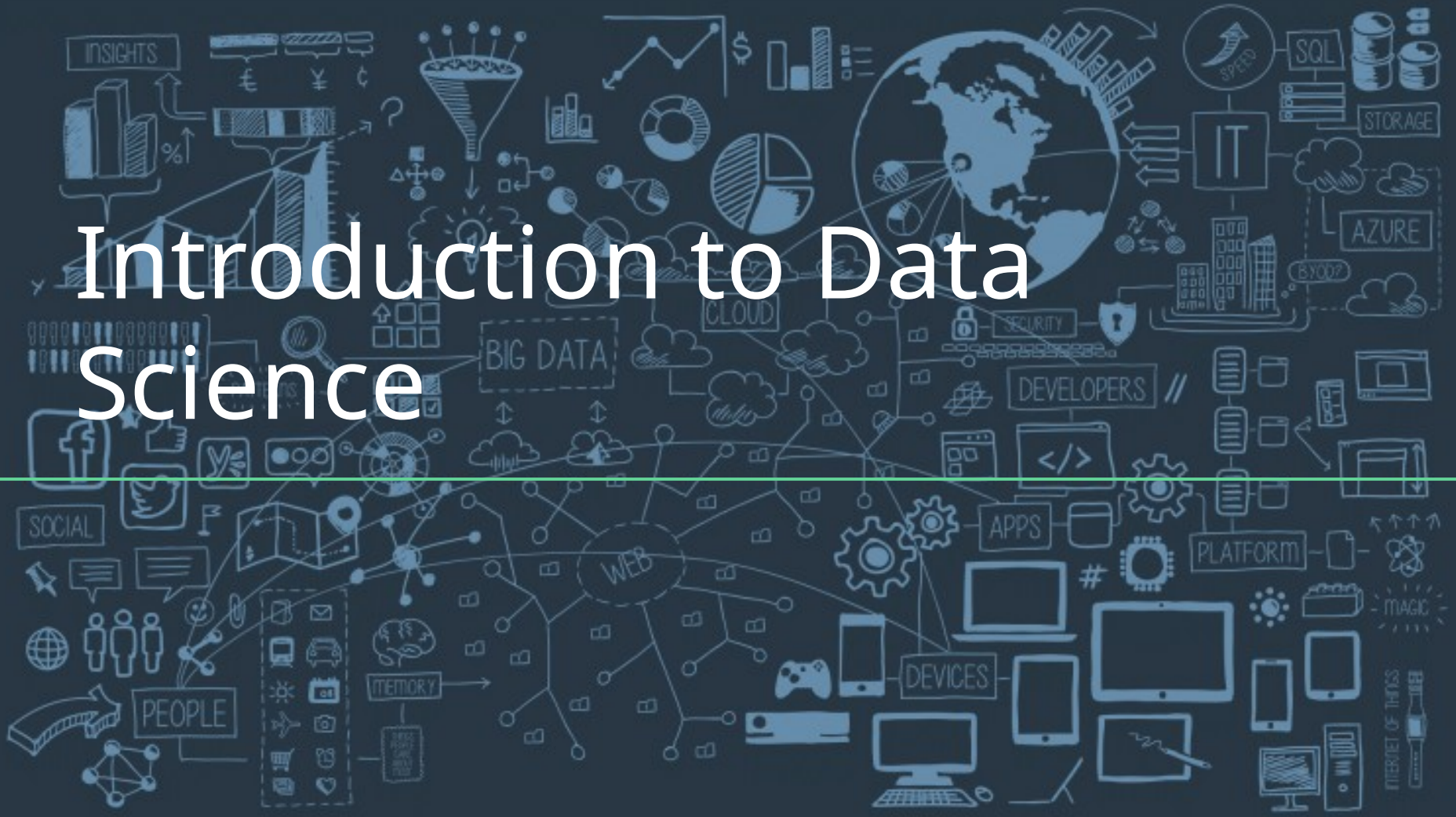
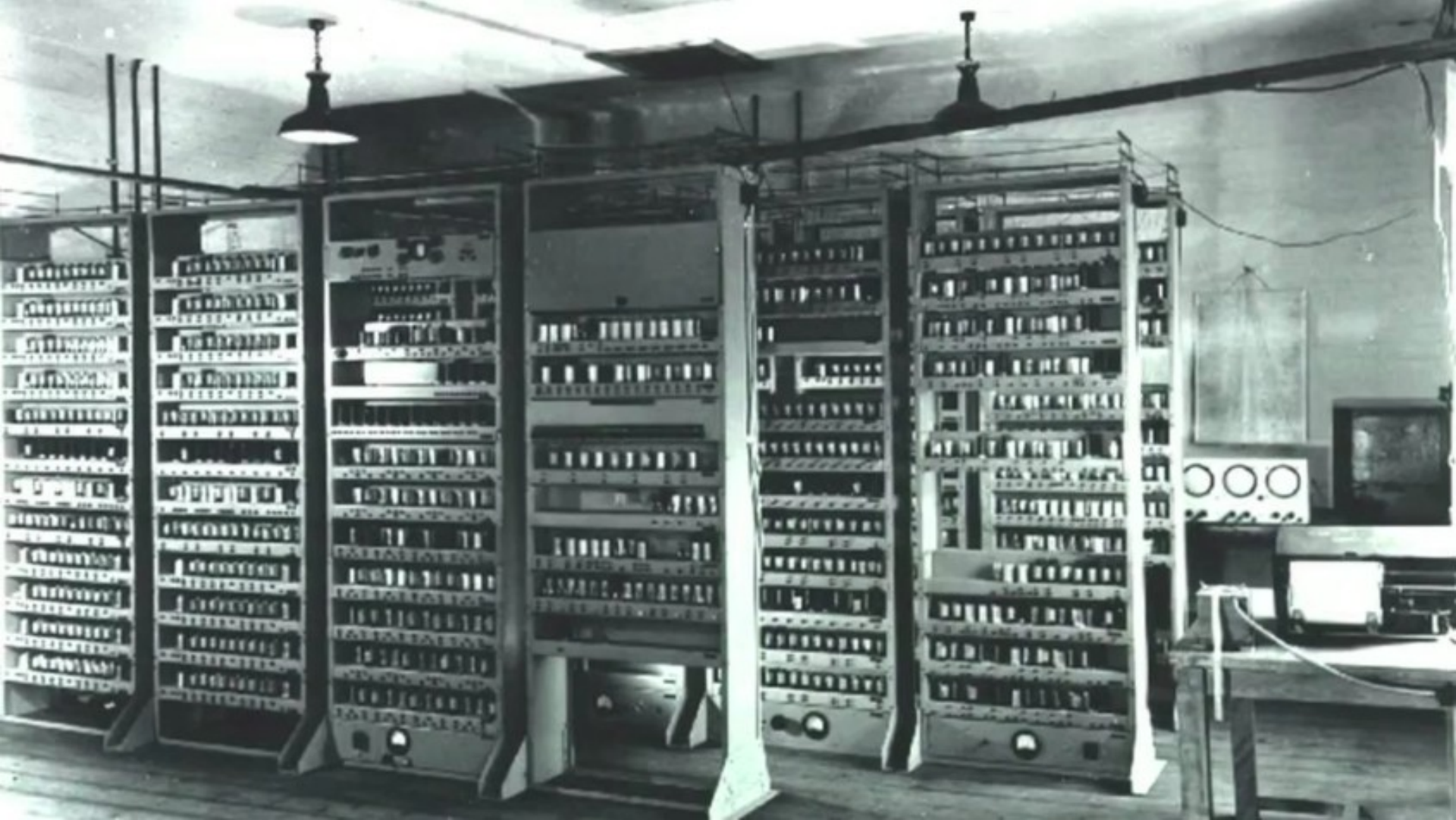
Introduction to Data Science









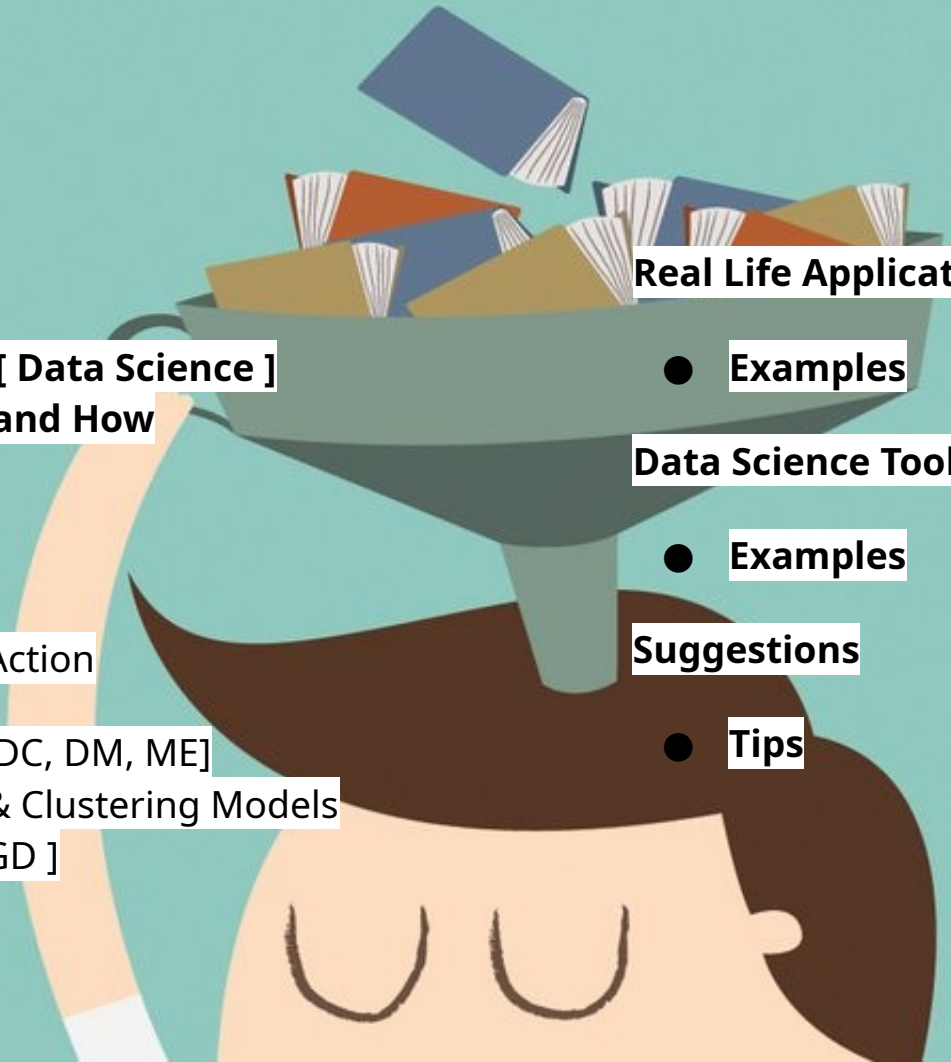


Data Science

1952 - Tic Tac Toe ⇒ Human vs Computer
1997 - Deep Blue - Chess ⇒ Exploring Solution Space
2011 - Watson - Jeopardy ⇒ Constructive Reasoning
2017 - AlphaGo - Go ⇒ Developing Intuition

In AlphaGo, no. of possibilities > total no. atoms in this universe.

Plan



Introduction

- **Definitions [Data Science]**
- **What, Why and How**
- **Examples**

Data Science - In Action

- Stages [DG, DC, DM, ME]
- Regression & Clustering Models
- Basics [LR, GD]

Real Life Application

- **Examples**

Data Science Tools

- **Examples**

Suggestions

- **Tips**

What is Data Science?

What is Data Science?

da•ta

Factual information, especially information organized for analysis or used to reason or make decisions.

Computer Science Numerical or other information represented in a form suitable for processing by computer.

Values derived from scientific experiments.

sci·ence (sī′əns)

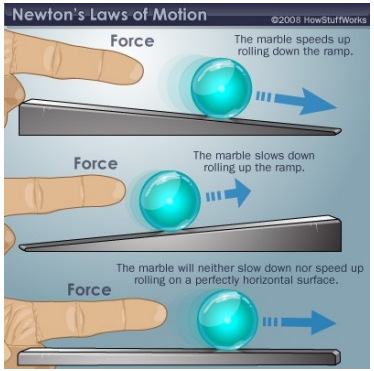
The **observation, identification, description, experimental investigation, and theoretical explanation of phenomena.** Ex. New advances in science and technology.

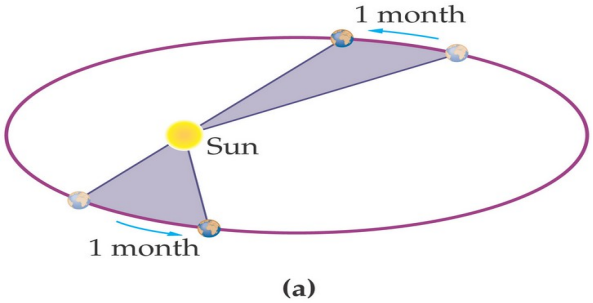
Such activities restricted to a class of natural phenomena. Ex. The science of astronomy.

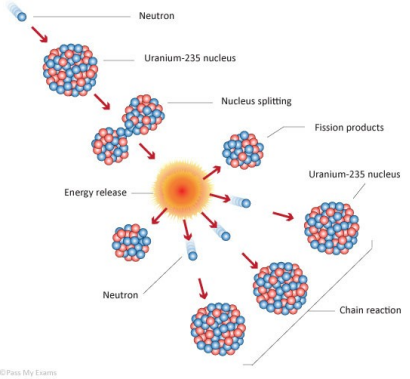
A systematic method or **body of knowledge in a given area.** Ex. The science of marketing.

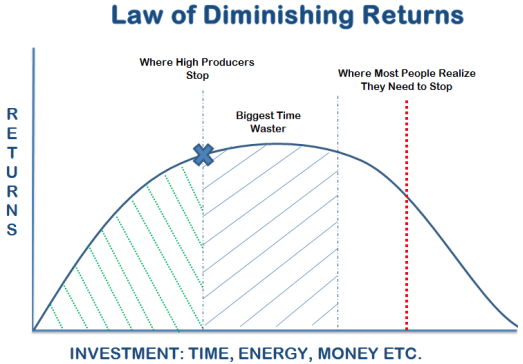
Archaic Knowledge, especially that gained through **experience**.

Data Science Examples

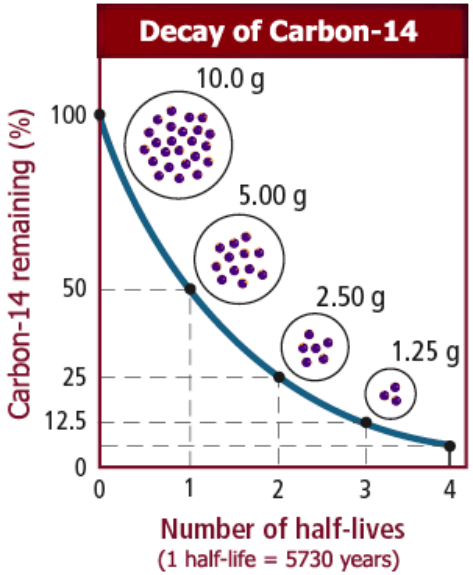












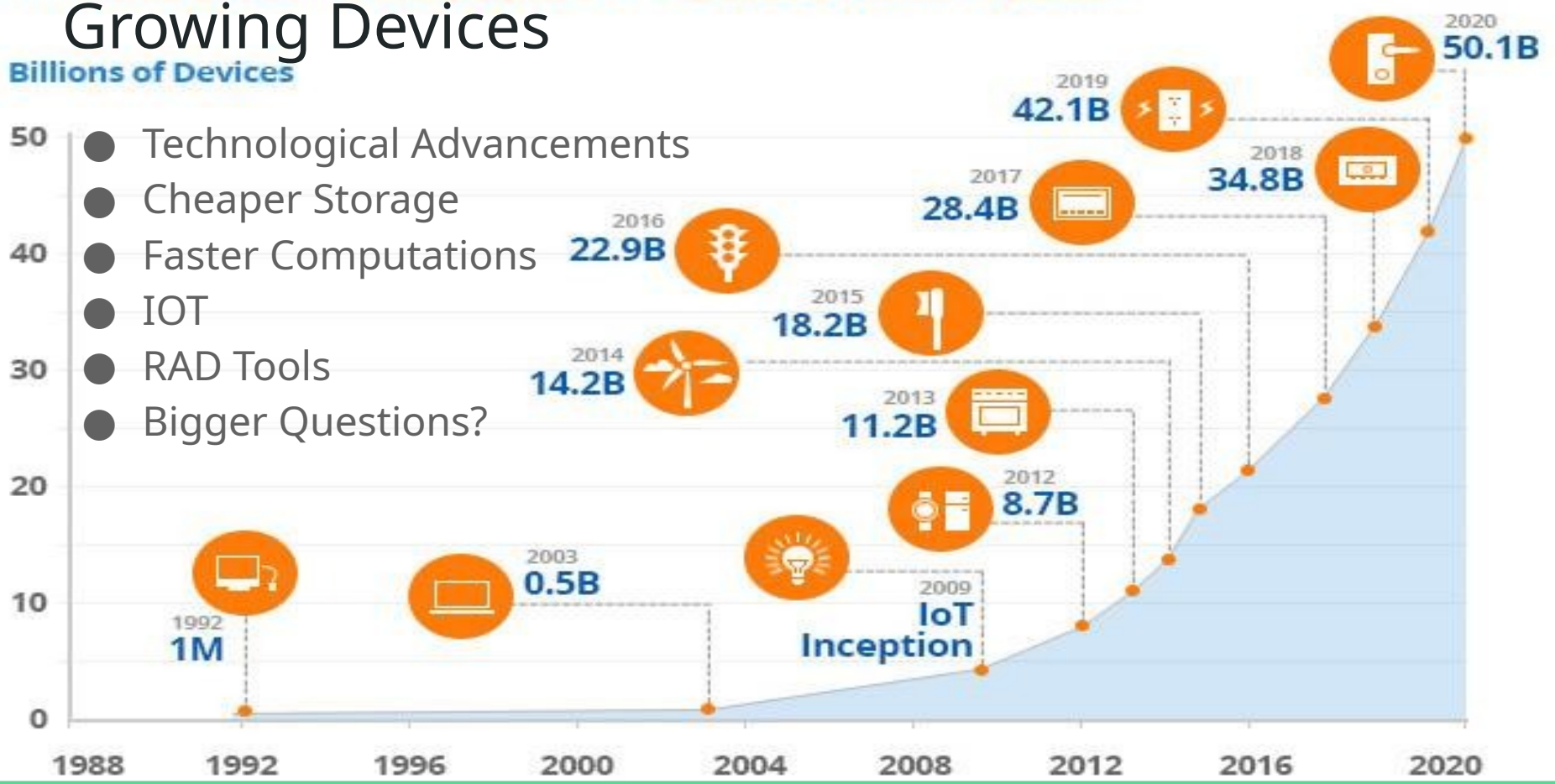
Why Data Science?

Growth in the internet of things

The number of connected devices will exceed 50 billion by 2020

Growing Devices

- Technological Advancements
- Cheaper Storage
- Faster Computations
- IOT
- RAD Tools
- Bigger Questions?



Information Explosion & Doubling Processing Power

Metcalfe's law states that the value of a telecommunications network is proportional to the square of the number of connected users of the system (n2).

Moore's law is the observation that the number of transistors in a dense integrated circuit doubles approximately every two years.

(Population - Thanks to Advanced Medical Sciences & Improving Health Care.)

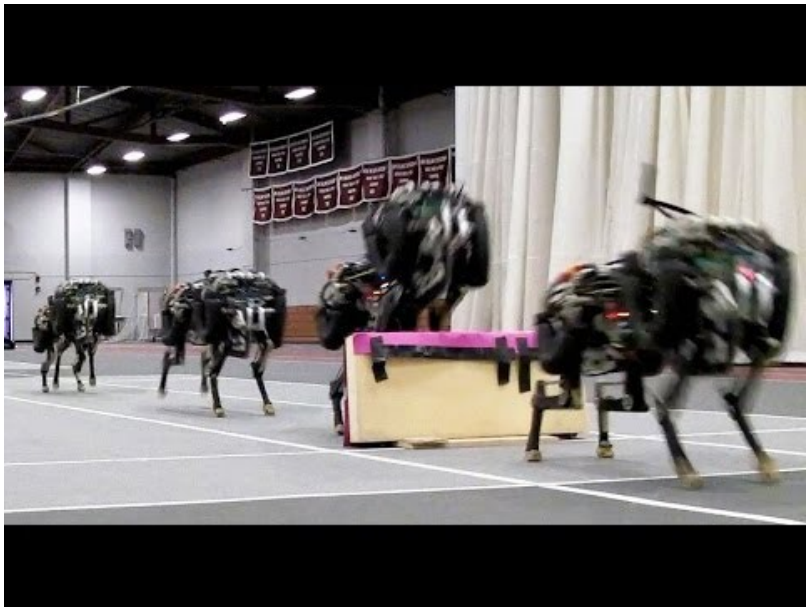
Sources: Wikipedia

How to do Data Science?

How to Data Science? - AI, ML



Rosey, Spacely, Jetson



MIT Cheetah Robot

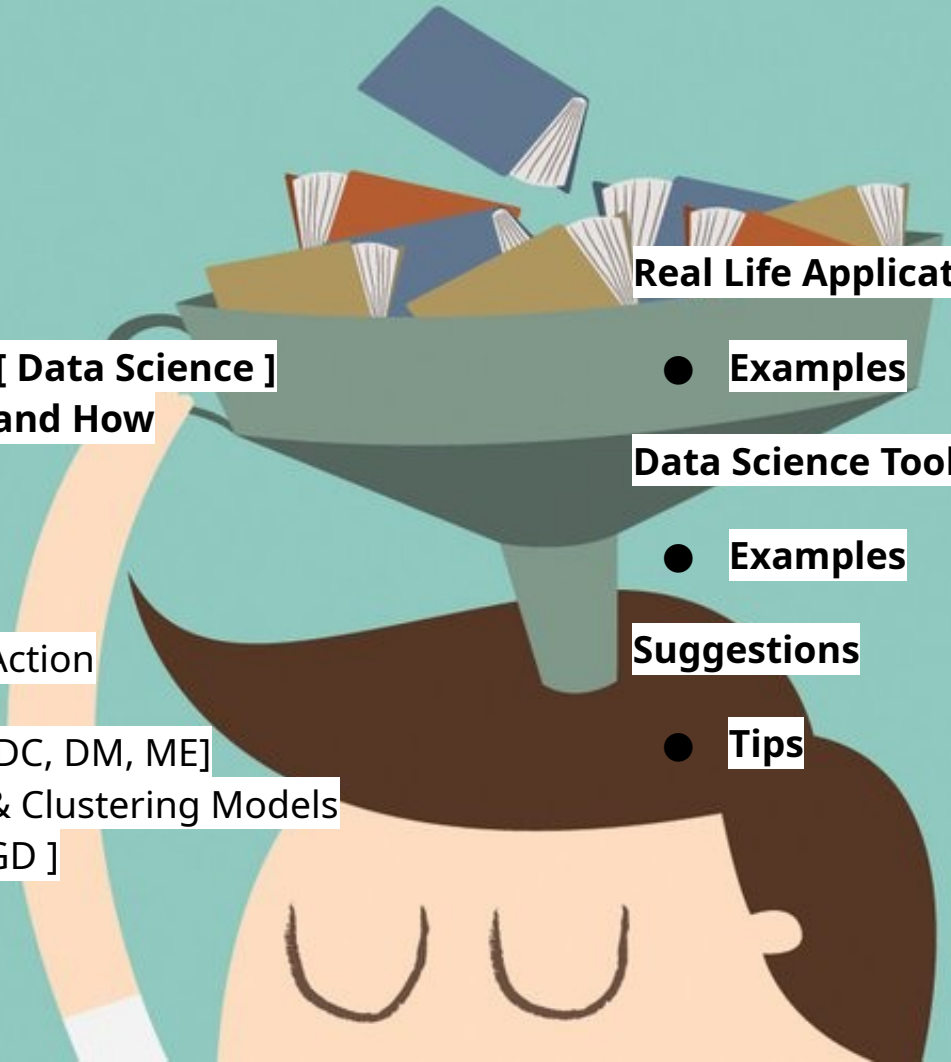
How to do Data Science

You can use lots of sophisticated analytical & Business Intelligent tools and come to a simple understandable explanations.

(or)

You can also use, simple tools like calculators or excel sheet to generate simple and simple results.

Plan



Introduction

- **Definitions [Data Science]**
- **What, Why and How**
- **Examples**

Data Science - In Action

- Stages [DG, DC, DM, ME]
- Regression & Clustering Models
- Basics [LR, GD]

Real Life Application

- **Examples**

Data Science Tools

- **Examples**

Suggestions

- **Tips**

Data Science - In Action

Stages of Data Science

- Purpose
- Relevant Data Collection

- Wrangling(cleansing)*
- Data Analytics
- Feature Engg.*
- Data Modelling*
- Data Prediction*
- Evaluation*

- Reportings
- Finalising Report

- Data Product Building (software development)
  - Architecture
  - Development
  - Testing
  - Deployment

(*) ⇒ Repetitive stages

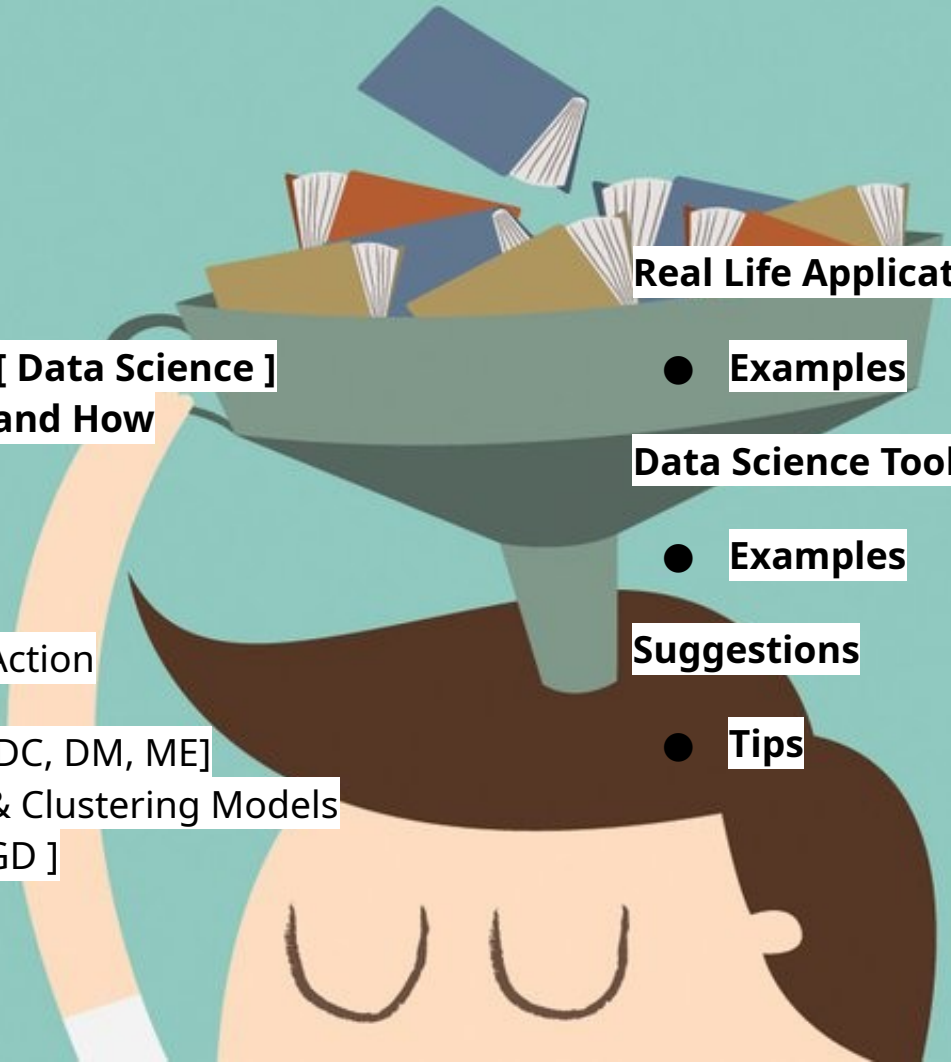
Data Model

- Random Forest Model
  - Bagging
- SVM
  - Linear Equation

Iris Dataset - Goal

<< Ipython Notebook >>

Plan



Introduction

- **Definitions [Data Science]**
- **What, Why and How**
- **Examples**

Data Science - In Action

- Stages [DG, DC, DM, ME]
- Regression & Clustering Models
- Basics [LR, GD]

Real Life Application

- **Examples**

Data Science Tools

- **Examples**

Suggestions

- **Tips**

Data Science - Real Life App

Few applications that inspired me

Passive Designs + AI



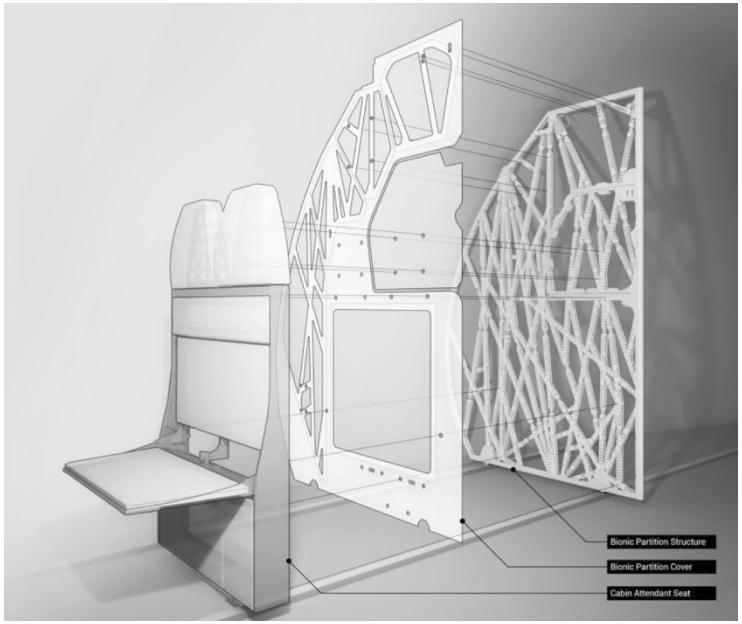
Maurice Cont

Director of Applied Research & Innovation

Autodesk, San Francisco Bay Area.

TED Talk: The incredible inventions of intuitive AI

Generative Designs > Passive Designs



AI Designed Lightweight Cabin Partition
Airbus - A320



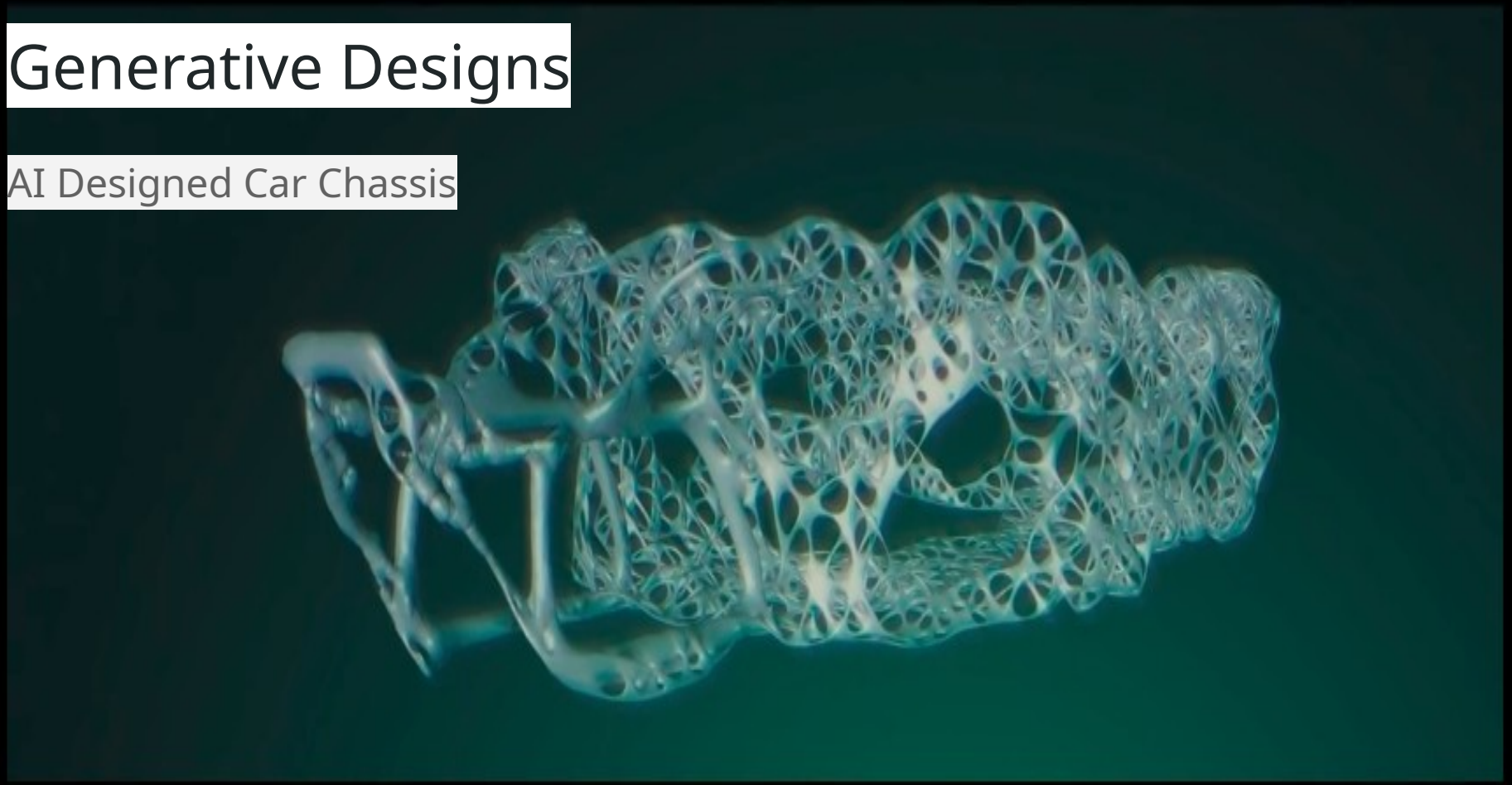
AI Designed Lightweight Drone Chassis

Generative Designs



Generative Designs

AI Designed Car Chassis



Music XRay

- Jimmy Lloyd Songwriter Showcase
- Popular songs share Melody & Rhythm
- Genere - 70
- Cluster 60
- Singer & Song Writer NY
- http://www.heidimerrill.com/epk/index.html



Pred Pole

- 2011 Santa Cruz Pred Pole
- Crime, Location & Date-Time
- https://www.predpol.com/

Results:

- 50% Crime Rate control
- 20% reduction in Crime Rate



Generative Designs

Project - Interlace





Plan

Introduction

- Definitions [Data Science]
- What, Why and How
- Examples

Data Science - In Action

- Stages [DG, DC, DM, ME]
- Regression & Clustering Models
- Basics [LR, GD]

Real Life Application

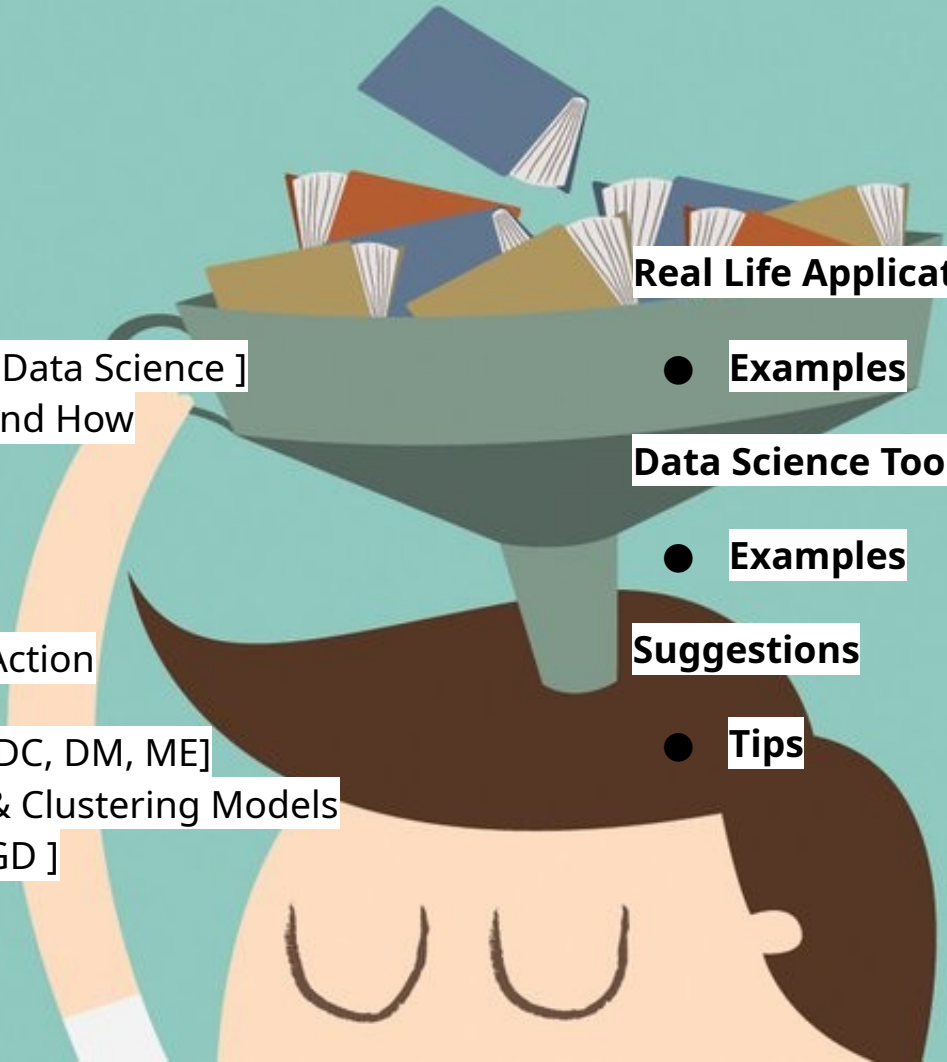
- **Examples**

Data Science Tools

- **Examples**

Suggestions

- **Tips**



Data Science - Tools

Too many to name, but none of them are close perfection.

Data Science Tools

- Languages: Scala, R, **Python**, Java, C#
- Lib: **Scikit**, DeepNet, **Tensor flow**, Theano, H20
- Frameworks: Apache Spark

These are some used by used us (Imaginea Labs - Data Sciences - 4th Floor, Hyd).

Suggestions

Challenges in DS & Tips to who want to start.

Suggestions?

- Data Preparation
  - “**Give me six hours to chop down a tree and I will spend the first four sharpening the axe**”. Abraham Lincoln
  - Python, Scala, Excel, Databases(regex).
- Data Analytics
  - “**Seeing is believing**”
  - Python(Matplotlib, Seaborn), D3.Js, Excel.
- Data Models
  - “**There are no perfect solutions, but some work better**”
  - Learn 2-3 types of Clustering, Regression Models(LR,RF,SVM,KNN,XGB)
- Evaluation
  - “**A product not tested is broken by default**”
  - Accuracy, RMSE, Precision-Recall, F1 Score

Questions?

Thank you

